

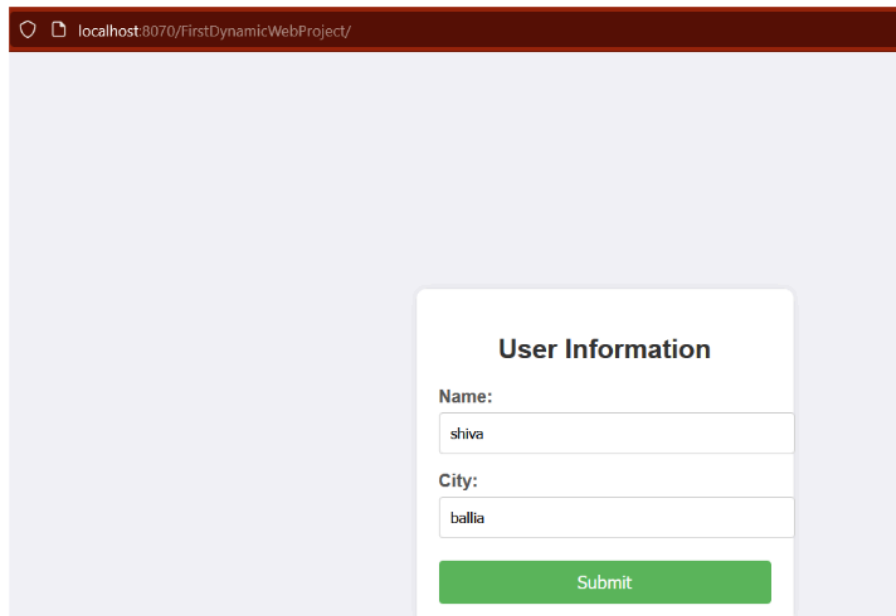
doGet() vs doPost()

In servlets, doGet() and doPost() are two key methods that handle HTTP GET and POST requests, respectively. Understanding the differences and use cases for each is crucial for web application development.

👉 doGet() Method

Register:

`http://localhost:8070/FirstDynamicWebProject/`



A screenshot of a web browser window. The address bar shows 'localhost:8070/FirstDynamicWebProject/'. The main content area has a light blue background. In the center, there is a white rounded rectangle titled 'User Information'. Inside this rectangle, there are two text input fields. The first is labeled 'Name:' and contains the text 'shiva'. The second is labeled 'City:' and contains the text 'ballia'. Below these fields is a green button with the text 'Submit'.

Output:

`http://localhost:8070/FirstDynamicWebProject/servletapp?uname=shiva&ucity=ballia`



A screenshot of a web browser window. The address bar shows 'localhost:8070/FirstDynamicWebProject/servletapp?uname=shiva&ucity=ballia'. Below the address bar, the text 'Name is : shiva' and 'City is : ballia' is displayed.

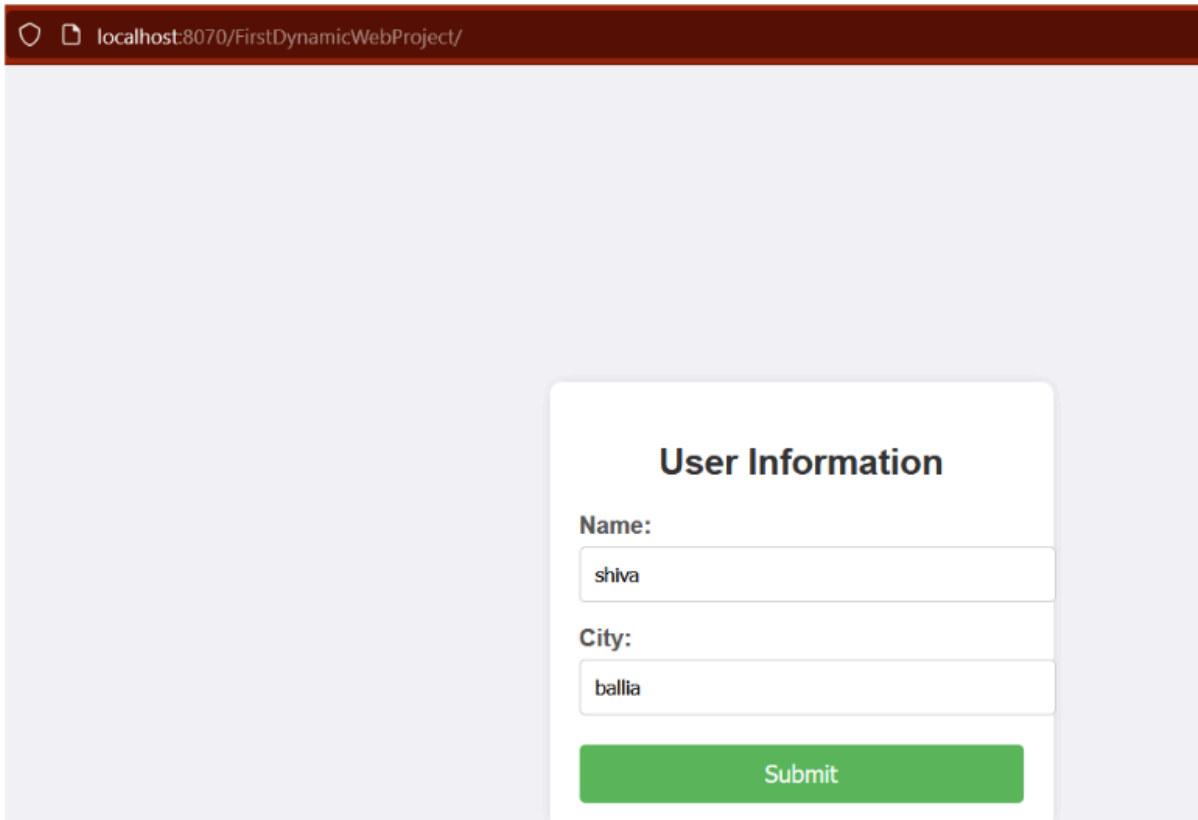
- **Purpose:** Handles HTTP GET requests, typically used to retrieve data from the server.
- **Signature:**

```
protected void doGet(HttpServletRequest req, HttpServletResponse res) throws ServletException, IOException
```

- **Characteristics:**
 - **Request Parameters:** Data is sent via the URL query string (e.g., ?param1=value1¶m2=value2).
 - **Data Length:** Limited by the maximum URL length (varies by browser, typically around 2048 characters).
 - **Caching:** Responses to GET requests can be cached by the browser or intermediary servers.
- **Typical Use Cases:**
 - Retrieving data (e.g., displaying a webpage, fetching a list of items).
 - Performing searches based on URL parameters.
 - Navigation actions where no server-side data changes occur.

👉 doPost() Method

`http://localhost:8070/FirstDynamicWebProject/`



localhost:8070/FirstDynamicWebProject/

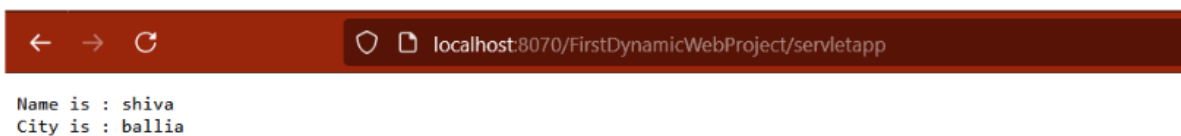
User Information

Name:

City:

Output:

`http://localhost:8070/FirstDynamicWebProject/servletapp`



← → ↻ localhost:8070/FirstDynamicWebProject/servletapp

Name is : shiva
City is : ballia

- **Purpose:** Handles HTTP POST requests, generally used to send data to the server for processing (e.g., form submissions).
- **Signature:**

```
protected void doPost(HttpServletRequest req,
```

HttpServletResponse res) **throws** ServletException, IOException

- **Characteristics:**

- **Request Parameters:** Data is sent in the request body, not the URL, allowing for larger amounts of data.
- **Data Length:** Not limited by the URL, suitable for submitting large datasets or files.
- **Security:** More secure than GET because data is not exposed in the URL (though still needs SSL for proper security).

- **Typical Use Cases:**

- Submitting form data (e.g., user registration, login forms).
- Uploading files to the server.
- Making changes to server data (e.g., creating, updating, or deleting records).

👉 How doGet() and doPost() method handle in service() method?

```
public void service(HttpServletRequest req, HttpServletResponse res)
    throws ServletException, IOException {
    String method = req.getMethod();

    if (method.equals("GET")) {
        doGet(req, res);
    } else if (method.equals("POST")) {
        doPost(req, res);
    } else if (method.equals("PUT")) {
        doPut(req, res);
    } else if (method.equals("DELETE")) {
        doDelete(req, res);
    } else {
        // Handle other methods or throw an exception
    }
}
```

```
    throw new ServletException("Unsupported method: " + method);  
  }  
}
```